

The logarithmic loss in the bounded-support tail estimate is sharp

Candidate full solution for arXiv:1706.07539

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Status. Candidate full solution, subject to human verification. The answer to the open question following (3.19b) is $\boxed{\gamma + 1}$. The published upper estimate already has this exponent; the construction below shows that no smaller exponent works under the paper’s hypotheses, even when the operator is positive, linear, and contractive on every L^p , $1 \leq p \leq b$.

Abstract

Ostrovsky and Sirota prove that a tail bound $T_f(y) \leq y^{-b}(\log y)^\gamma L(\log y)$ is transferred by an operator of equal-power type to a bound with logarithmic exponent $\gamma + 1$, and ask for the optimal replacement of that exponent. We build a positive linear L^p -contraction which, on sparse atomic blocks, maps a function having the source tail to an image whose tail attains the $\gamma + 1$ power along an unbounded sequence. Thus the one-logarithm loss is unavoidable in the stated operator class.

1 Source question

The source is E. Ostrovsky and L. Sirota, *Maximal and other operators in exponential Orlicz and Grand Lebesgue Spaces*, arXiv:1706.07539 (2017) [1]. In Example 3.2, pp. 17–18, they fix $b > 1$, $\gamma > -1$, and a positive slowly varying function L , assume

$$T_f(y) \leq y^{-b}(\log y)^\gamma L(\log y), \quad y \geq e,$$

and derive

$$T_g(y) \leq C y^{-b}(\log y)^{\gamma+1} L(\log y).$$

Immediately after (3.19b), they ask: “what is the ultimate value instead $\gamma + 1$ in the last estimate?”

2 Answer and proof intuition

The exponent is $\gamma + 1$. A moment estimate alone suggests the extra logarithm, but does not prove it is genuinely attained. The key is to make it attained on sparse scales.

On block n , put N_n equal-mass input atoms at successively smaller heights. Their b -th powers are proportional to $1/k$. A norm-one functional on $\ell_{N_n}^b$ sends this particular input vector to its full ℓ^b -norm on one output atom of the same mass. Since $\sum_{k \leq N_n} 1/k \sim \log N_n$, the output height gains exactly one logarithm in its b -th power. Hölder’s inequality shows that the resulting block operator is contractive not only on L^b , but simultaneously on all L^p , $1 \leq p \leq b$. Rapid separation of the blocks prevents their input tails from interfering, while the output atoms witness sharpness along a sequence.

3 Formal sharpness theorem

For a measurable random variable h , write $T_h(y) = \mathbb{P}(|h| > y)$.

$$T_g(y) \leq T^{(m,L)}(y/C_3) \quad (3.15a)$$

or equivalently

$$\sup_{p \geq 1} \left[\frac{|g|_p}{\psi_{m,L}(p)} \right] < \infty. \quad (3.15b)$$

Example 3.2. The case of bounded support.

This case is more complicated. Recall the following notation for tail function

$$T^{<b,\gamma,L>}(x) \stackrel{\text{def}}{=} x^{-b} (\ln x)^\gamma L(\ln x), \quad x \geq e,$$

where as before $L = L(x)$, $x \geq 1$ is the positive continuous slowly varying function as $x \rightarrow \infty$, and let as before

$$b = \text{const} \in (1, \infty), \quad \gamma = \text{const} > -1.$$

and recall also notation for the following correspondent $\Psi(b)$ function

$$\psi^{<b,\gamma,L>}(p) \stackrel{\text{def}}{=} C_1(b, \gamma, L) (b-p)^{-(\gamma+1)/b} L^{1/b} \left(\frac{1}{b-p} \right), \quad 1 \leq p < b.$$

Let the source (measurable) function $f(\cdot)$ be such that

$$\|f\| \in G\psi^{<b,\gamma,L>} \iff \|f\| G\psi^{<b,\gamma,L>} < \infty, \quad (3.16)$$

then also $\|g\| \in G\psi^{<b,\gamma,L>}$ and moreover

$$\|g\| G\psi^{<b,\gamma,L>} \leq K^{b,\gamma}(\lambda) \|f\| G\psi^{<b,\gamma,L>}. \quad (3.17)$$

But if we assume the following tail restriction on the function f

$$T_f(y) \leq T^{<b,\gamma,L>}(y), \quad y \geq e, \quad (3.18)$$

then we conclude only

$$T_g(y) \leq C_5(b, \gamma, L) y^{-b} (\ln y)^{\gamma+1} L(\ln y), \quad y \geq e \quad (3.19a)$$

Figure 1: Source paper, p. 17: assumptions (3.18) and upper estimate (3.19a).

or equally

$$T_g(y) \leq T^{<b, \gamma+1, L>}(y/C_6), \quad y \geq C_6 e. \quad (3.19b)$$

Open question: what is the ultimate value instead “ $\gamma+1$ ” in the last estimate?

4. 4. MAIN RESULT. THE CASE OF DIFFERENT POWERS.

Let as before some function $\psi = \psi(p)$, $p \in [1, b]$, $b = \text{const} \in (1, \infty]$ from the set $\Psi(b)$ be a given. Suppose in this section that in the inequalities (1.1) or (1.2) $f \in G\psi$, $\|f\|G\psi < \infty$, $\lambda > \nu \geq 0$, and denote $\Delta = \lambda - \nu$; ($\Delta > 0$),

$$\zeta(p) = \zeta[\psi, \Delta](p) := p^\Delta \psi(p). \quad (4.0)$$

Obviously, $\zeta(\cdot) \in \Psi(b)$.

Let for beginning $\|f\|G\psi = 1$, then $|f|_p \leq \psi(p)$, $p \in [1, b]$. We deduce from the inequality (1.1) taking into account the estimate $|f|_p \leq \psi(p)$, $p \in [1, b]$ alike the foregoing section denoting $g = Q[f]$

$$|g|_p \leq Z \left[\frac{p}{p-1} \right]^\nu p^\Delta \psi(p) = Z \left[\frac{p}{p-1} \right]^\nu \zeta[\psi, \Delta](p). \quad (4.1)$$

It follows immediately from proposition (3.1) or theorem 3.1

$$\|g\|G\zeta[\psi, \Delta] \leq Z K_\nu[\zeta[\psi, \Delta]](\zeta, b). \quad (4.2)$$

We proved in fact the following result.

Theorem 4.1. Suppose as above that the function $f(\cdot)$ belongs to the space $G\psi$ for some generating function ψ from the set $\Psi(b)$, $1 < b \leq \infty$. Let in (1.1) $\lambda > \nu \geq 0$. Our statement: the function $g = Q[f]$ from the relations (1.1) belongs to the *other* certain Grand Lebesgue Space $G\zeta$, or equivalently to the correspondent exponential Orlicz space $L(M_\zeta)$:

$$\|g\|G\zeta[\psi, \Delta] \leq Z K_\nu(\zeta, b) \|f\|G\psi, \quad (4.3)$$

Figure 2: Source paper, p. 18: (3.19b) and the open question.

Theorem 1 (sharpness of (3.19)). *Let $b > 1$, $\gamma > -1$, and let $L : (0, \infty) \rightarrow (0, \infty)$ be continuous and slowly varying at infinity. There are a probability space, a nonnegative measurable function f , and a positive linear operator Q such that*

1. $\|Qh\|_p \leq \|h\|_p$ for every $1 \leq p \leq b$;
2. after a fixed scalar normalization of f ,

$$T_f(y) \leq y^{-b}(\log y)^\gamma L(\log y), \quad y \geq e;$$

3. for $g = Qf$ and every $\beta < \gamma + 1$,

$$\limsup_{y \rightarrow \infty} \frac{T_g(y)}{y^{-b}(\log y)^\beta L(\log y)} = \infty.$$

More precisely, for some $y_n \rightarrow \infty$,

$$T_g(y_n) \gtrsim y_n^{-b}(\log y_n)^{\gamma+1} L(\log y_n).$$

Consequently $Q \in \text{Type}(\lambda, \lambda)$ with $Z(Q) = 1$ for every $\lambda \geq 0$, and the exponent $\gamma + 1$ in (3.19) cannot be replaced by any smaller exponent in the class considered in [1].

Proof. Set

$$\ell(t) = t^\gamma L(t), \quad F(x) = x^{-b} \ell(\log x).$$

The function ℓ is regularly varying with index γ , while F is regularly varying with index $-b$. We use only the uniform convergence and Potter bounds for regularly varying functions [2, Theorems 1.2.1 and 1.5.6].

Fix $0 < \alpha < b$. Choose a sufficiently rapidly increasing sequence $R_n \rightarrow \infty$, and put

$$w_n = e^{-bR_n}, \quad N_n = \lfloor e^{\alpha R_n} \rfloor.$$

It may be chosen so that

$$\sum_n (N_n + 1)w_n < 1, \text{ and } \sum_{m > n} N_m w_m \leq w_n, \quad (1)$$

and so that every height in block $n + 1$ is larger than every height in block n . Indeed, $N_n w_n \asymp e^{-(b-\alpha)R_n}$, whereas the smallest height introduced below grows like $e^{(b-\alpha)R_n/b}$ up to a subexponential factor. Thus both requirements follow recursively by taking R_{n+1} large enough.

Let $c_0 > 0$ be fixed and define, for $1 \leq k \leq N_n$,

$$z_{n,k}^b = \frac{c_0 \ell(R_n)}{k w_n}. \quad (2)$$

These heights decrease with k . Uniformly in $k \leq N_n$,

$$\frac{\log z_{n,k}}{R_n} = \frac{1}{b} \left(b - \frac{\log k}{R_n} + o(1) \right) \in \left[\frac{b-\alpha}{b} + o(1), 1 + o(1) \right].$$

The uniform convergence theorem therefore gives constants $0 < c < C < \infty$ such that, for all large n and all $k \leq N_n$,

$$c, k w_n \leq F(z_{n,k}) \leq C, k w_n. \quad (3)$$

Construct an atomic probability space with input atoms $A_{n,k}$ ($1 \leq k \leq N_n$) and one output atom B_n , every one of these atoms having mass w_n . Add one residual atom carrying the unused mass. Define $f_0(A_{n,k}) = z_{n,k}$ and let f_0 vanish on the output and residual atoms.

We first verify the input tail. At a level between $z_{n,k+1}$ and $z_{n,k}$, at most kw_n of block n lies above the level, and all later blocks contribute at most w_n by (1). Equation (3), the bounded ratio $z_{n,k}/z_{n,k+1} = ((k+1)/k)^{1/b}$, and Potter's bound for F show

$$T_{f_0}(y) \leq C_1 F(y), \quad y \geq y_0.$$

The same Potter bound handles the gaps between successive blocks; choosing the first block sufficiently far out handles $e \leq y < y_0$. Finally replace f_0 by $f = af_0$, where $a > 0$ is sufficiently small. Since F is regularly varying with negative index, this fixed dilation absorbs C_1 and yields

$$T_f(y) \leq F(y), \quad y \geq e. \quad (4)$$

For each block set

$$Y_n = \left(\sum_{k=1}^{N_n} z_{n,k}^b \right)^{1/b}, \quad a_{n,k} = \frac{z_{n,k}^{b-1}}{Y_n^{b-1}}.$$

Define the positive linear operator Q by

$$Qh(B_n) = \sum_{k=1}^{N_n} a_{n,k} h(A_{n,k}), \quad Qh = 0 \quad \text{off } \bigcup_n B_n.$$

If $b' = b/(b-1)$, then

$$\sum_{k=1}^{N_n} a_{n,k}^{b'} = 1.$$

For $1 < p \leq b$, one has $p' \geq b'$, hence $\|(a_{n,k})\|_{p'} \leq \|(a_{n,k})\|_{b'} = 1$. Hölder's inequality, followed by summation over the disjoint blocks, gives

$$\|Qh\|_p^p = \sum_n w_n \left| \sum_k a_{n,k} h(A_{n,k}) \right|^p \leq \sum_{n,k} w_n |h(A_{n,k})|^p \leq \|h\|_p^p.$$

The case $p = 1$ follows from $\max_k a_{n,k} \leq 1$. This proves the simultaneous contraction assertion.

The coefficients were chosen to norm the particular input vector:

$$Qf_0(B_n) = \sum_k \frac{z_{n,k}^b}{Y_n^{b-1}} = Y_n.$$

By (2) and the harmonic sum,

$$\begin{aligned} Y_n^b &= \frac{c_0 \ell(R_n)}{w_n} \sum_{k=1}^{N_n} \frac{1}{k} \\ &\asymp w_n^{-1} R_n^{\gamma+1} L(R_n). \end{aligned}$$

Because $\log L(R_n) = o(\log R_n)$, this implies $\log Y_n = R_n + o(R_n)$. Slow variation now yields

$$w_n \asymp Y_n^{-b} (\log Y_n)^{\gamma+1} L(\log Y_n). \quad (5)$$

For $g = Qf = aQf_0$, the atom B_n gives $T_g(aY_n/2) \geq w_n$. Combining this with (5) and fixed-dilation invariance of regular variation, for every $\beta < \gamma + 1$,

$$\frac{T_g(aY_n/2)}{(aY_n/2)^{-b} (\log(aY_n/2))^\beta L(\log(aY_n/2))} \gtrsim (\log Y_n)^{\gamma+1-\beta} \longrightarrow \infty.$$

This proves all assertions. $\square$

4 Why this answers the source question

The paper already proves the upper estimate with exponent $\gamma + 1$. Theorem 1 shows that every smaller exponent fails, for each $b > 1$ and $\gamma > -1$, and even for positive linear contractions. Such a contraction satisfies

$$\|Qh\|_p \leq \|h\|_p \leq \left(\frac{p}{p-1}\right)^\lambda \|h\|_p, \quad 1 < p < b,$$

for every $\lambda \geq 0$. It therefore lies in every equal-power class $\text{Type}(\lambda, \lambda)$ used in Section 3 of the source. The optimal universal exponent is consequently exactly $\gamma + 1$.

5 Verification notes

The proof is entirely analytic. The accompanying script `code/check_finite_blocks.py` checks finite blocks for representative parameters. It verifies numerically that the coefficient vector has $\ell^{b'}$ -norm one, its $\ell^{p'}$ -norm is at most one for $p \leq b$, and the output scale contains the harmonic factor. This is a consistency check, not part of the proof. The command

```
python code/check_finite_blocks.py
```

was run for $b = 2$, $\gamma = 0.5$, $L = 1$, $\alpha = 0.8$, and $R = 4, 6, 8, 10, 12$. In every case $\|a\|_{b'} = 1$ to twelve displayed decimal places and all tested $\|a\|_{p'} \leq 1$; the normalized $\gamma + 1$ output scale stayed between 0.66 and 0.70, while the ratio for the smaller test exponent $\beta = \gamma + 0.5$ increased from 1.51 to 2.56.

The most important human-review points are:

1. the uniform use of regular variation in (3);
2. the tail bookkeeping across rapidly separated blocks;
3. the endpoint norm comparison $p' \geq b'$, which is what makes the operator simultaneously contractive for all $p \leq b$.

6 Novelty check and limitations

A bounded search was performed on 9 August 2026. The local run registry, solution, attempt, and proof-gap indexes were searched for arXiv:1706.07539, the paper title, and the core tail/exponent terms. Web searches used the exact phrase “what is the ultimate value”, the exact title, and combinations of “bounded support”, “Grand Lebesgue”, “maximal operator”, and the $\gamma + 1$ tail exponent. They surfaced the source paper but no later arXiv paper or primary-source answer to this exact question. This is a bounded novelty check, not a definitive citation survey.

The result identifies the sharp exponent for the broad operator class defined only through the L^p estimates in the source. A narrower class with extra structure (for example, a particular geometric maximal operator) could obey a better endpoint tail law; no such narrower claim is made here.

References

- [1] E. Ostrovsky and L. Sirota, *Maximal and other operators in exponential Orlicz and Grand Lebesgue Spaces*, arXiv:1706.07539, 2017.
- [2] N. H. Bingham, C. M. Goldie, and J. L. Teugels, *Regular Variation*, Encyclopedia of Mathematics and its Applications 27, Cambridge University Press, 1987.